

Plants Need Sunlight!

Grade 2 Science • Hands-on Discovery Lesson Plan

Welcome to an exciting science adventure! This lesson helps Grade 2 students explore exactly why plants need light to live, grow, and stay healthy. Through a simple, fun 7-day experiment with moong seeds, children become real scientists—predicting, observing, and drawing conclusions.

What We Will Learn Today

- **Discover:** Why sunlight is like healthy food for green plants.
- **Predict:** What happens to a plant when it is hidden away in the dark.
- **Experiment:** Set up a fair test using two real live plant pots.
- **Observe:** Watch, track, and draw your very own plants for one week!

Fun Fun Analogy for Kids!

Just like how you need a yummy plate of rice, lentils, or rotis to run and play, a plant needs **sunlight** to make its own energy! Sunlight is the plant's kitchen.

Real-Life Plant Friends Around Us

Have you seen these famous plants around your neighborhood or home?

- **Tulsi (Holy Basil):** Look closely! Grandparents always place the Tulsi pot outside on a sunny balcony or next to a big bright window. Now you know why!
- **Money Plant:** Have you noticed how a money plant inside the room always bends and leans over towards the glass window? It is stretching its leaves to catch the sunbeams!
- **Sunflowers:** True to their name, these bright yellow flowers literally turn their heads to follow the sun across the sky all day long.



The Great Light Experiment

Let's find out: Does a plant grow better with or without sunlight?



Science Toolkit (Materials Needed)



10-12 Moong Seeds (Mung beans from the kitchen)



2 Small Pots or Cups (Labelled Pot A and Pot B)



Potting Soil (A few scoops of dirt)



1 Cardboard Box (A dark shoebox with a tight lid)



Water & Small Cup (To give them equal drinks)



Observation Worksheet & Colored Crayons



Step-by-Step Guide

Step 1: Prep the Pots

Fill both of your cups with the exact same amount of soft soil. Push 2 or 3 moong seeds gently into the dirt (about as deep as your fingernail). Cover them gently.

Step 2: Give a Equal Drink

Give both pots the exact same amount of water (half a small cup is perfect). We want everything to be perfectly equal!

Step 3: Place Your Pots

Pot A (Sunlight): Put this pot on a bright, sunny windowsill.

Pot B (Dark): Hide this pot completely inside the dark cardboard box and shut the lid tightly so no light sneaks in.

Step 4: Care and Track

Every single day at the same time, give both pots an equal spoonful of water. Peek inside the box quickly to see what's happening, then close it right back up!

Scientist Secret: The Variable

In this experiment, we keep the soil, water, and pots exactly the same. The only thing different is the **Sunlight!** This is what scientists call a *variable*.



My Science Observation Diary

Name: Class:

..... Date:

.....



My Predictions (What I Think Will Happen)

I think Pot A (Sunlight) will look:

I think Pot B (Dark) will look:

Draw Pot A Prediction

Draw Pot B Prediction



7-Day Growth Tracker

Write down how the leaves look (Green? Yellow?), how tall they are, and if they look strong or droopy!

Day	Pot A (Sunlight Window)	Pot B (Dark Closed Box)
Day 1		
Day 3		
Day 5		
Day 7		



My Ultimate Scientific Conclusion

After 7 days of careful observation, I discovered that:

Pot A grew up to be and

Pot B grew up to be and

This proves that plants absolutely need to grow healthy and strong!



Word Bank & Fun Quiz

Let's test our green-thumb knowledge!



Science Vocabulary Flashcards



Sunlight

The bright golden light from the sun. Plants use it like fuel to build energy and food!



Experiment

A super fun test we do to see if something is true by changing just one thing.



Observe

To watch something very closely with your eyes and take notes like a real detective.



Healthy vs. Weak

Healthy: Green, tall, and straight.

Weak: Pale yellow, short, and very droopy.



Quick Pop Quiz!

Q1. What are the four things a plant needs to survive?

Answer: Plants need sunlight, fresh water, good soil, and clean air.

Q2. What happens to a plant if you lock it away from sunlight?

Answer: Without sunlight, it becomes sad and weak, its leaves turn pale yellow, and its stem droops down.

Q3. Why did Pot A grow so much better than Pot B?

Answer: Pot A had plenty of sunlight to create food and energy, while Pot B was left hungry in the dark!

Q4. What is the special scientific word for a plant leaning toward light?

Answer: It's a big bonus word: **Phototropism!** (Photo = Light, Tropism = Turning).



Home Assignment Challenge

Go around your house today and find 3 different plants. Ask a family member where each plant gets its daily sunlight beams!